Periodontal (Gum) Disease: Prevention & Management

What is Periodontal (Gum) Disease? Periodontal disease is an inflammatory infection caused by bacteria in plaque biofilm. It begins as *gingivitis* (gum inflammation), which causes redness and bleeding but no bone loss. If plaque continues unchecked, it progresses to *periodontitis*, where the gums pull away, pockets form, and bone is lost. Without treatment, teeth can become loose and ultimately be lost. WHO describes it as a chronic infection that destroys tooth-supporting structures. In the U.S., roughly 40% of adults have periodontitis, a figure that rises with age. The first steps to control it are daily hygiene and professional cleanings.

Causes and Risk Factors: Periodontal disease is triggered by **plaque**, a sticky film of bacteria. The host immune response to plaque causes inflammation and tissue damage. Major risk factors include:

- *Poor Oral Hygiene:* Infrequent brushing/flossing allows plaque to harden into calculus, fueling infection.
- *Smoking/Tobacco:* Tobacco use greatly increases risk and severity of gum disease, and impairs healing after treatment. (WHO and ADA note tobacco is strongly linked to periodontal destruction.)
- *Diabetes:* Uncontrolled diabetes impairs immune response; conversely, periodontitis can worsen glycemic control. It's a bidirectional relationship.
- *Genetics & Stress:* Some people are more susceptible due to genetic factors or stress.
- *Medication and Illness:* Certain meds (anticonvulsants, immunosuppressants) or conditions (HIV) can affect gum health.
- *Diet and Nutrition:* Diets high in sugar contribute to overall oral disease. WHO advises balanced nutrition to help prevent periodontal disease.

A **table** below summarizes indications:

Factor	Effect on Gums	Prevention/Action
Plaque buildup	Initiates inflammation and infection	Brush 2×/day, floss daily, use antiseptic mouthwash
Tobacco	Major risk, delays healing	Quit smoking (ask for help); mention risks in counseling
Diabetes	Increases infection severity	Monitor glucose; see dentist regularly; emphasize hygiene
Poor nutrition	Weaker immune response	Eat balanced diet; vitamin C supports healing
Dry mouth (e.g., from meds)	Reduces natural cleansing by saliva	Stay hydrated; saliva substitutes if needed

Prevention and Home Care: Key to gum health is removing plaque **daily**. The CDC advises brushing teeth twice daily for 2 minutes with fluoride toothpaste and flossing at least once. WHO recommends cleaning teeth with fluoride toothpaste twice daily to prevent plaque buildup. Other advice:

- Use a soft-bristled toothbrush and replace it every 3 months. Electric toothbrushes can improve plaque removal.
- Floss or use interdental cleaners daily, especially between molars where plaque hides.
- Rinse with an antimicrobial mouthwash if recommended by a dentist.
- Stop smoking and avoid tobacco entirely. Even chewing tobacco causes severe gum disease.
- Maintain a balanced diet low in added sugars. Eat crunchy fruits/veggies to naturally clean teeth.
- Schedule professional cleanings (oral prophylaxis) every 6 months or as advised. During these visits, hygienists scale (remove) plaque and tartar even in periodontal pockets.

Treatment Options:

- **Non-Surgical Therapy (First-line):** Scaling and root planing (deep cleaning) to remove plaque and calculus from teeth and roots. This is the cornerstone of therapy. ADA guidelines conclude that "scaling and root planing without adjuncts is the treatment of choice for patients who have periodontitis". Patients also receive oral hygiene instruction. Often multiple visits are needed to clean all areas.
- **Antibiotic Adjuncts:** For moderate to severe cases or patients with systemic risks, dentists may prescribe antiseptic rinses (e.g. chlorhexidine) or systemic antibiotics (like doxycycline) as adjunctive therapy. ADA notes that sub-antimicrobial dose doxycycline may be used with SRP in some cases.
- **Surgical Therapy:** If deep pockets persist (typically >6mm) after initial therapy, periodontal surgery may be indicated to access roots or reshape bone. Procedures include flap surgery (lifting gums to clean roots), bone grafts for regeneration, and gum grafts for recession. Surgery reduces pocket depths so patients can maintain better hygiene.
- **Maintenance:** After active therapy, patients enter periodontal maintenance: professional cleanings every 3 months (instead of 6) to control plaque and prevent recurrence. Failure to maintain leads to rapid relapse.